A PROJECT REPORT

On

"Z-TEST PERFORMANCE COMPARISON SIMULATOR"

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Abstract

This document describes the Z-Test Performance Comparison Simulator, an interactive web tool written in Python with the Streamlit library. The tool compares the average performance of two groups using a two-sample Z-test. The user types the raw scores of each group. The program calculates the sample means and sample sizes from those scores, runs the test, and shows the result as a clear verdict with charts. Two simulation experiments are included. The first repeats the test thousands of times to estimate how often it rejects the null hypothesis. The second shows how this rate changes as the sample size grows. The tool is meant for learning and for demonstrating how the Z-test behaves.

1. Introduction

1.1 Problem statement

When two groups have different averages, it is not obvious whether the difference is real or caused by random variation. A statistical hypothesis test answers this question. Students often learn the formula but do not see how the result changes when the data, the spread or the sample size changes. This project builds a tool that makes these effects visible.

1.2 Objectives

- Accept raw data for two groups and calculate the mean and sample size of each group from that data.
- Perform a two-sample, two-sided Z-test and report the Z-score, p-value, critical value and decision.
- Show the result visually so it is easy to read.
- Estimate the rejection rate (power or Type I error rate) by Monte Carlo simulation.
- Show the effect of sample size on the rejection rate.

1.3 Scope

The tool performs a two-sided Z-test for two independent groups when the population standard deviations are known. It does not perform t-tests or one-sided tests.

2. Theoretical background

2.1 Hypotheses

- Null hypothesis H_0: the two population means are equal ($\mu_1 = \mu_2$).
- Alternative hypothesis H_1: the two population means are different ($\mu_1 \neq \mu_2$).

2.2 Test statistic

The Z-score measures the gap between the two sample means in units of its standard error:

$$Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$$

Here $\bar{x}$ is the sample mean, σ is the known population standard deviation and n is the sample size of each group.

2.3 Assumptions

- The population standard deviation of each group is known.
- The two samples are independent.
- The sample means are approximately normally distributed. This holds when the populations are normal, or when the samples are large enough for the central limit theorem to apply.

2.4 p-value and critical value

- Two-sided p-value: $p = 2 \times P(Z > |z|)$, calculated from the standard normal distribution.
- Critical value: $z_{crit} = \Phi^{-1}(1 - \alpha/2)$. For $\alpha = 0.05$ this is 1.96.

2.5 Decision rule

If $p < \alpha$ (equivalently $|z| > z_{crit}$), reject H_0 and conclude that the groups differ significantly. Otherwise, fail to reject H_0. Failing to reject does not prove that the groups are equal. It only means the evidence is not strong enough.

2.6 Error types and power

Term	Meaning
Type I error	Rejecting H_0 when it is true (a false alarm). Its probability equals α.
Type II error	Failing to reject H_0 when it is false (a missed difference).
Power	The probability of correctly rejecting H_0 when it is false. Power = 1 − P(Type II error).

3. System overview

3.1 Technology

Component	Purpose
Python 3	Programming language
Streamlit	Web interface and interaction
NumPy	Arrays, averages and random sampling
SciPy	Normal distribution functions (p-value, critical value)
Matplotlib	Charts

3.2 Inputs

Input	Description
Group 1 scores	Raw numbers typed or pasted by the user
Group 2 scores	Raw numbers typed or pasted by the user
Population std of each group	Known true spread of each group
Alpha	Significance level (0.01, 0.05 or 0.10)

3.3 Outputs

- Number of values and average of each group, calculated from the data.
- Z-score, p-value, critical value and the decision (reject or fail to reject H_0).
- A verdict bar showing where the Z-score falls compared with the rejection zones.
- Charts of the two sampling distributions and of the standard normal curve.
- Results of the repeated simulation and of the sample size experiment.

3.4 Workflow

1. The user enters the scores, the population standard deviations and alpha.
2. The user clicks Calculate.
3. The program calculates the means and sample sizes, runs the test and shows the verdict.
4. The user can open the tabs “Repeat it many times” and “Change the sample size” to run the simulations.

4. Implementation

4.1 Reading the data

The typed text is split at commas, semicolons, spaces and new lines, and each piece is converted to a number. The program stops with a clear message if the text contains anything that is not a number, or if a group has fewer than two valid values.

4.2 Calculations

The mean of each group is the sum of its values divided by the number of values. The sample size is the number of values. The test itself is one function:

```
def ztest(m1, m2, s1, s2, n1, n2, alpha):
    z = (m1 - m2) / math.sqrt(s1**2 / n1 + s2**2 / n2)
    p = 2 * norm.sf(abs(z))
    critical_z = norm.ppf(1 - alpha / 2)
    return z, p, critical_z
```

4.3 Verdict display

A bar shows the number line from −4 to 4. The two ends are shaded as rejection zones and the middle as the zone where a gap could be luck. A marker shows the observed Z-score. A headline states the decision in plain words.

4.4 Charts for the single test

- Sampling distributions: each group's sample mean is drawn as a normal curve with mean $\bar{x}$ and standard error $\sigma/\sqrt{n}$. A large overlap means the gap is easy to get by chance.
- Standard normal curve: shows the distribution of Z when H_0 is true, the rejection region and the observed Z-score.

4.5 Repeated simulation (Monte Carlo)

The two calculated means are treated as the true population means. In each repeat the program draws n random values for each group from a normal distribution with the given mean and standard deviation, calculates the two sample means and the Z-score, and checks whether H_0 is rejected. The share of repeats that reject H_0 is reported.

- If the true means differ, this share estimates the power of the test.
- If the true means are equal, this share estimates the Type I error rate and should be close to α.

All repeats are generated together with NumPy, which makes the simulation fast:

```
def simulate_z(mu1, mu2, s1, s2, n1, n2, sims):
    a = np.random.normal(mu1, s1, (sims, n1)).mean(axis=1)
    b = np.random.normal(mu2, s2, (sims, n2)).mean(axis=1)
    return (a - b) / np.sqrt(s1**2 / n1 + s2**2 / n2)
```

4.6 Sample size experiment

The simulation is repeated for sample sizes 10, 20, 30 and so on up to a chosen maximum, using the same sample size for both groups. The rejection rate is plotted against the sample size. Reference lines mark α and the common 80% power target.

4.7 Handling of state

Streamlit runs the whole script again after every click or change. The inputs are placed in a form with one Calculate button, and the calculated values and simulation results are kept in Streamlit's session state so that they stay on the screen while the user explores the tabs.

5. Worked example

The program starts with the following example values.

Item	Group 1	Group 2
Number of values (n)	10	10
Sample mean	0.455	0.401
Population std (σ)	0.10	0.12

Calculation with $\alpha = 0.05$:

- Standard error = $\sqrt{(0.10^2/10 + 0.12^2/10)} = \sqrt{0.00244} \approx 0.0494$
- $Z = (0.455 - 0.401) / 0.0494 \approx 1.09$
- p-value ≈ 0.274
- Critical value = ±1.96

Because 0.274 is greater than 0.05 (and $|Z| = 1.09$ is less than 1.96), the test fails to reject H_0. With only 10 values per group, a gap of 0.054 could easily be caused by chance. Entering more data or data with a larger gap changes the verdict.

6. Verification

The statistical functions were checked separately from the web interface, using 20,000 simulated repeats for each case (n = 50 per group, σ = 0.10 and 0.12, α = 0.05).

Case	Expected	Observed
Equal true means (0.40 and 0.40)	Rejection rate close to α = 5%	5.2%
Different true means (0.45 and 0.40)	Power about 62% from theory	62.3%

The example in Section 5 was also calculated independently with SciPy and gave the same Z-score (1.093) and p-value (0.2743) as the formulas above. These checks confirm the calculations. The visual layout was checked by looking at the running application.

7. How to run the program

7.1 Requirements

Python 3 and the libraries Streamlit, SciPy, NumPy and Matplotlib.

```
pip install streamlit scipy numpy matplotlib
```

7.2 Files

```
z_test/
  z_test_simulator_ui.py
  .streamlit/
    config.toml
```

The file config.toml is optional. It sets a light colour theme.

7.3 Start the application

```
streamlit run z_test_simulator_ui.py
```

The application opens in the web browser, usually at http://localhost:8501. Press Ctrl + C in the terminal to stop it.

8. Limitations

- The Z-test needs the population standard deviations to be known. In most real studies they are unknown and a t-test is used instead.
- With small samples the result is reliable only if the populations are approximately normal.
- Only a two-sided test for two independent groups is supported.
- The sample size experiment uses the same sample size for both groups.
- Simulation results change slightly every time because they use random numbers.
- Data must be typed or pasted by hand.

9. Possible future improvements

- Add a t-test option that estimates the standard deviation from the data.
- Add upload of data from a CSV file.
- Add one-sided tests, confidence intervals and an effect size measure.
- Allow different sample sizes in the sample size experiment.
- Add an option to fix the random seed so that results can be repeated exactly.

10. Conclusion

The simulator turns the Z-test from a formula into something the user can see and experiment with. By entering real data, the user sees how the means, the Z-score and the decision are obtained. The simulations then show why a single test result can be misleading and how sample size, spread and the size of the real gap affect the chance of detecting a difference.

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